The Fundamental Theorem of Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Part 2: evaluating definite integrals

Evaluate each definite integral:

a $\int_1^3 (2x+1)\,dx$ $[x^2+x]_1^3 = 12-2 = 10.$

b $\int_0^{\pi} \sin x\,dx$ $[-\cos x]_0^{\pi} = 1-(-1) = 2.$

c $\int_1^4 \sqrt{x}\,dx$ $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{16}{3}-\frac{2}{3} = \frac{14}{3}.$

(a) $[x^2+x]_1^3 = 12-2 = 10.$; (b) $[-\cos x]_0^{\pi} = 1-(-1) = 2.$; (c) $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{16}{3}-\frac{2}{3} = \frac{14}{3}.$

2 [Foundational] • Target: Part 2: evaluating definite integrals

Evaluate $\int_0^1 e^{2x}\,dx$, to three decimal places.

$\left[\frac{1}{2}e^{2x}\right]_0^1 = \frac{1}{2}(e^2-1) \approx 3.195.$

3 [Foundational] • Target: Part 1: differentiating an integral

Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx}\int_2^x (t^3+1)\,dt$.

$= x^3+1$ (directly, by FTC Part 1).

4 [Foundational] • Target: Part 1: differentiating an integral

Find $\frac{d}{dx}\int_1^{x^2} \sin(t)\,dt$ using the chain rule with FTC.

$= \sin(x^2)\cdot 2x = 2x\sin(x^2).$

5 [Foundational] • Target: Part 1: differentiating an integral

Let $g(x) = \int_0^x \sqrt{4+t^2}\,dt$. Find $g'(3)$.

$g'(x) = \sqrt{4+x^2}$, so $g'(3) = \sqrt{13}$.

6 [Foundational] • Target: Average value and applications

Find the average value of $f(x) = x^2$ on $[0, 3]$.

$\frac{1}{3}\int_0^3 x^2\,dx = \frac{1}{3}\cdot 9 = 3.$

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7 [Foundational] • Target: Average value and applications

A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \le t \le 4$. Find the total distance traveled, and the average velocity over this interval.

Distance $= \int_0^4 (3t^2 + 2)\,dt = [t^3 + 2t]_0^4 = 64 + 8 = 72$ m. Average velocity $= \frac{72}{4} = 18$ m/s.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Further FTC practice

Evaluate $\int_0^2 (x^3 - 2x + 3)\,dx$.

$\left[\frac{x^4}{4} - x^2 + 3x\right]_0^2 = (4 - 4 + 6) - 0 = 6.$

9 [Intermediate] • Target: Further FTC practice

Find $\frac{d}{dx}\int_x^5 \cos(t^2)\,dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.

$\int_x^5 \cos(t^2)\,dt = -\int_5^x \cos(t^2)\,dt$, so the derivative is $-\cos(x^2)$.

10 [Intermediate] • Target: Further FTC practice

Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.

$\frac{1}{\pi}\int_0^\pi \sin x\,dx = \frac{1}{\pi}[-\cos x]_0^\pi = \frac{2}{\pi} \approx 0.637.$

11 [Intermediate] • Target: Mean Value Theorem for integrals

Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).

Average value $= \frac{1}{3}\int_0^3 x^2\,dx = 3$ (found earlier). Solve $c^2 = 3$: $c = \sqrt{3}$ (taking the positive root within $[0, 3]$).

12 [Intermediate] • Target: Mean Value Theorem for integrals

Evaluate $\int_2^5 f(x)\,dx$ given that the average value of f on $[2, 5]$ is 7.

Average value $= \frac{1}{5-2}\int_2^5 f(x)\,dx = 7$, so $\int_2^5 f(x)\,dx = 7(3) = 21$.

13 [Intermediate] • Target: Integrals of piecewise functions

Evaluate $\int_0^4 f(x)\,dx$ where $f(x) = x$ for $x \le 2$ and $f(x) = 2x - 2$ for $x > 2$.

$\int_0^2 x\,dx + \int_2^4 (2x - 2)\,dx = \left[\frac{x^2}{2}\right]_0^2 + [x^2 - 2x]_2^4 = 2 + (8 - 0) = 10.$

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14 [Intermediate] • Target: Visualizing the two parts of FTC together

For $g(x) = \int_0^x (t^2 - 2)\,dt$, find $g'(x)$ using FTC Part 1, then find where g has a local minimum by analyzing the sign of $g'(x)$.

$g'(x) = x^2 - 2$ (FTC Part 1). Setting $g'(x) = 0$: $x = \pm\sqrt{2}$. Since g' changes from negative to positive at $x = \sqrt{2}$ (for $x > 0$), g has a local minimum at $x = \sqrt{2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

The rate of water flowing into a tank is $r(t) = 5 + 2\sin(t)$ liters/minute. Find the total volume that flows in during $[0, \pi]$ minutes, using FTC.

Volume $= \int_0^{\pi} (5 + 2\sin t)dt = [5t - 2\cos t]_0^{\pi} = (5\pi + 2) - (0 - 2) = 5\pi + 4 \approx 19.71$ liters.

16 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 15.01 Double Integrals

Find $\frac{d}{dx}\int_0^{x^3} \sqrt{1 + t^2}\,dt$ using the chain rule with FTC.

$= \sqrt{1 + (x^3)^2} \cdot 3x^2 = 3x^2\sqrt{1 + x^6}$.

17 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Find $\frac{d}{dx}\int_{x^2}^5 \cos(t)\,dt$, using $\int_{x^2}^5 = -\int_5^{x^2}$ before differentiating.

$\int_{x^2}^5 \cos t\,dt = -\int_5^{x^2} \cos t\,dt$, so the derivative is $-\cos(x^2) \cdot 2x = -2x\cos(x^2)$.

18 [Challenging] • Target: Net change and the FTC • Cross-Lesson: 2.01 Evaluating Limits

A tank starts with 200 liters of water. Water flows out at rate $r(t) = 3t + 2$ liters/minute for the first 5 minutes. Use FTC to find the amount of water in the tank at $t = 5$.

Net change $= -\int_0^5 (3t + 2)\,dt = -\left[\frac{3t^2}{2} + 2t\right]_0^5 = -(37.5 + 10) = -47.5$ liters. Amount remaining $= 200 - 47.5 = 152.5$ liters.

19 [Challenging] • Target: Visualizing FTC Part 2 as area • Cross-Lesson: 2.02 Continuity

The graph shows $f(x) = x^2 - 1$ on $[0, 3]$. Use FTC to find $\int_0^3 (x^2 - 1)\,dx$ and explain why the result is not simply the area of the shaded region (part of the curve dips below the axis).

$\left[\frac{x^3}{3} - x\right]_0^3 = (9 - 3) - 0 = 6$. This is the SIGNED area: the small region below the axis on $[0, 1]$ is subtracted from the larger region above the axis on $[1, 3]$, so 6 is a net value, not the total unsigned area.

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20 [Challenging] • Target: Combining FTC Part 1 and Part 2 in one problem • Cross-Lesson: 3.01 The Derivative from First Principles

Let $F(x) = \int_1^x (3t^2 - 2)\,dt$. Find $F(2)$ using Part 2, then verify $F'(2)$ using Part 1 and confirm it matches direct substitution into the integrand.

$F(2) = [t^3 - 2t]_1^2 = (8 - 4) - (1 - 2) = 4 - (-1) = 5$. By Part 1, $F'(x) = 3x^2 - 2$, so $F'(2) = 12 - 2 = 10$, which matches the integrand $3t^2 - 2$ evaluated at $t = 2$: $3(4) - 2 = 10$ ✓.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: FTC Part 1 (Derivative of an Accumulation Function)

If $g(x) = \int_1^x \sqrt{t^3 + 1}\,dt$, find $g'(x)$.

a. $\sqrt{x^3 + 1}$ ✓

b. $\frac{3x^2}{2\sqrt{x^3 + 1}}$

c. $\sqrt{x^3 + 1} - \sqrt{2}$

d. $\frac{1}{3}(x^3 + 1)^{3/2}$

By FTC Part 1, $\frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x)$. Thus $g'(x) = \sqrt{x^3 + 1}$.

22 [Foundational] • Target: FTC Part 1 (Derivative of an Accumulation Function)

Find $\frac{d}{dx}\left[\int_x^5 \cos(t^2)\,dt\right]$.

a. $-\cos(x^2)$ ✓

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\cos(25) - \cos(x^2)$

Reverse limits: $\int_x^5 = -\int_5^x$. Differentiating yields $-\cos(x^2)$.

23 [Foundational] • Target: FTC Part 1 (Derivative of an Accumulation Function)

Find $h'(x)$ for $h(x) = \int_0^{x^2} \sin(t)\,dt$.

a. $2x\sin(x^2)$ ✓

b. $\sin(x^2)$

c. $2x\cos(x^2)$

d. $\sin(x^4)$

Chain rule with FTC 1: $\frac{d}{dx}\left[\int_a^{u(x)} f(t)\,dt\right] = f(u(x)) \cdot u'(x) = \sin(x^2) \cdot 2x = 2x\sin(x^2)$.

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24 [Foundational] • Target: FTC Part 1 with Variable Limits on Both Bounds

Find $F'(x)$ for $F(x) = \int_{2x}^{3x} (t^2 + 1)\,dt$.

a. $27x^2 + 3 - 2(4x^2 + 1) = 19x^2 + 1$ ✓

b. $19x^2 + 1$

c. $27x^2 - 8x^2 = 19x^2$

d. $5x^2 + 1$

Split at 0: $\int_{2x}^{3x} = \int_{0}^{3x} - \int_{0}^{2x}$. Differentiating:
$3[(3x)^2 + 1] - 2[(2x)^2 + 1] = 3(9x^2 + 1) - 2(4x^2 + 1) = 27x^2 + 3 - 8x^2 - 2 = 19x^2 + 1$.

25 [Foundational] • Target: FTC Part 1 with Variable Limits on Both Bounds

Evaluate $\frac{d}{dx}\left[\int_{\sin x}^{\cos x} e^t\,dt\right]$.

a. $-e^{\cos x}\sin x - e^{\sin x}\cos x$ ✓

b. $e^{\cos x} - e^{\sin x}$

c. $e^{\cos x}\sin x + e^{\sin x}\cos x$

d. $-e^{\cos x} + e^{\sin x}$

FTC with chain rule: $e^{\cos x} \cdot (-\sin x) - e^{\sin x} \cdot (\cos x) = -e^{\cos x}\sin x - e^{\sin x}\cos x$.

26 [Foundational] • Target: FTC Part 1 with Variable Limits on Both Bounds

Find the inflection points of $g(x) = \int_{0}^{x} (t^3 - 3t)\,dt$.

a. $x = 1$ **and** $x = -1$ ✓

b. $x = 0$ and $x = \pm\sqrt{3}$

c. $x = 3$

d. No inflection points

$g'(x) = x^3 - 3x$. $g''(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \Rightarrow x = \pm 1$. Sign of $g''(x)$ changes at $x = -1$ and $x = 1$.

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27 [Foundational] • Target: FTC Part 2 (Evaluating Definite Integrals)

Evaluate $\int_1^3 (3x^2 - 4x + 1)\,dx$.

a. 12 ✓

b. 14

c. 10

d. 16

Antiderivative $F(x) = x^3 - 2x^2 + x$. $F(3) = 27 - 18 + 3 = 12$. $F(1) = 1 - 2 + 1 = 0$. Value $= 12 - 0 = 12$.

28 [Foundational] • Target: FTC Part 2 (Evaluating Definite Integrals)

Evaluate $\int_0^{\pi/4} \sec^2(x)\,dx$.

a. 1 ✓

b. $\sqrt{2}$

c. 1/2

d. 0

$[\tan x]_0^{\pi/4} = \tan(\pi/4) - \tan(0) = 1 - 0 = 1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: FTC Part 2 (Evaluating Definite Integrals)

Evaluate $\int_1^e \frac{4}{x}\,dx$.

a. 4 ✓

b. $4e$

c. $4\ln(e) - 4 = 0$

d. 1

$[4\ln|x|]_1^e = 4\ln(e) - 4\ln(1) = 4(1) - 4(0) = 4$.

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30 [Intermediate] • Target: Average Value of a Function

State the formula for the average value f_{avg} of a continuous function f on $[a, b]$.

a. $f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$ ✓

b. $f_{\text{avg}} = \frac{f(a)+f(b)}{2}$

c. $f_{\text{avg}} = \int_a^b f(x)\,dx$

d. $f_{\text{avg}} = \frac{f(b)-f(a)}{b-a}$

The average value is the total accumulation divided by the width of the interval: $\frac{1}{b-a}\int_a^b f(x)\,dx$.

31 [Intermediate] • Target: Average Value of a Function

Find the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 2]$.

a. 2 ✓

b. 4

c. 3

d. 1

$\frac{1}{2-0}\int_0^2 (3x^2 - 2x)\,dx = \frac{1}{2}[x^3 - x^2]_0^2 = \frac{1}{2}[8-4] = \frac{1}{2}(4) = 2.$

32 [Intermediate] • Target: Average Value of a Function

By the Mean Value Theorem for Integrals, find $c \in [0, 3]$ such that $f(c) = f_{\text{avg}}$ for $f(x) = 2x$.

a. 1.5 ✓

b. 1.0

c. 2.0

d. 0.75

$f_{\text{avg}} = \frac{1}{3}\int_0^3 2x\,dx = \frac{1}{3}[x^2]_0^3 = \frac{1}{3}(9) = 3$. Set $2c = 3 \Rightarrow c = 1.5$.

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33 **[Intermediate]** • **Target: FTC Part 1 (Derivative of an Accumulation Function)**

For $F(x) = \int_a^x f(t)dt$ with f continuous, the proof of FTC Part 1 uses the Mean Value Theorem for Integrals to write $F(x+h) - F(x)$ as:

a. hf'(x)

b. $f(c) \cdot h$ for some $c \in [x, x+h]$ ✓

c. $f(x) \cdot (x+h)$

d. f(x+h)-f(x)

MVT for integrals gives $\int_x^{x+h} f(t)dt = f(c)h$ for some c between x and $x+h$; letting $h \to 0$, $c \to x$.

34 **[Intermediate]** • **Target: FTC Part 1 with Variable Limits on Both Bounds**

If f has a jump discontinuity at $c \in (a, b)$, the accumulation function $F(x) = \int_a^x f(t)dt$ is still:

a. continuous at c (though not differentiable there) ✓

b. discontinuous at c

c. differentiable at c

d. undefined at c

Integrating a bounded (even discontinuous) function always yields a continuous accumulation function; differentiability can fail at the jump.

35 **[Intermediate]** • **Target: FTC Part 2 (Evaluating Definite Integrals)**

FTC Part 1: $G(x) = \int_0^x \sin(t^2)dt$. $G'(x) =$

a. $\sin(x^2)$ ✓

b. $\cos(x^2)$

c. $2x\sin(x^2)$

d. $\sin x^2/x$

Applying Fundamental Theorem of Calculus principles, FTC1: $G'(x) = \sin(x^2)$. Therefore, the verified correct option is $\sin(x^2)$.

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36 [Intermediate] • Target: Average Value of a Function

FTC Part 2: $\int_a^b f(x)dx = F(b) - F(a)$ where $F' = f$. Evaluate $\int_0^{\pi} \cos x\, dx$:

a. 0 ✓

b. 2

c. -2

d. 1

Applying Fundamental Theorem of Calculus principles, $[\sin x]_0^{\pi} = 0 - 0 = 0$. Therefore, the verified correct option is 0.

37 [Intermediate] • Target: Part 2: evaluating definite integrals

Net change theorem: $\int_1^4 v(t)dt$ gives the net displacement if $v(t)$ is velocity. For $v = 2t$:

a. 15 ✓

b. 8

c. 30

d. 4

Applying Fundamental Theorem of Calculus principles, $[t^2]_1^4 = 16 - 1 = 15$. Therefore, the verified correct option is 15.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Part 1: differentiating an integral • Cross-Lesson: 2.01 Evaluating Limits

If f is continuous on $[a, b]$ and F is any antiderivative, FTC Part 2 requires no assumption that F is:

a. equal to f

b. unique $--$ any antiderivative works, since they differ by a constant ✓

c. bounded on $[a, b]$

d. differentiable everywhere

Any two antiderivatives of f differ by a constant C, which cancels in $F(b) - F(a)$.

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39 [Challenging] • Target: Average value and applications • Cross-Lesson: 2.02 Continuity

The Net Change Theorem $\int_a^b F'(x)dx = F(b) - F(a)$ requires F' to be:

a. continuous and positive

b. integrable on $[a, b]$ ✓

c. a polynomial

d. bounded and increasing

The theorem (a restatement of FTC Part 2) only needs F' integrable on $[a, b]$.

40 [Challenging] • Target: Further FTC practice • Cross-Lesson: 3.01 The Derivative from First Principles

FTC Part 1 shows every continuous function has an:

a. **antiderivative** ✓

b. derivative everywhere

c. limit

d. inflection point

Constructs antiderivative $G(x) = \int_a^x f\,dt$.

41 [Challenging] • Target: Mean Value Theorem for integrals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Mean value theorem for integrals: $\int_a^b f\,dx = f(c)(b - a)$ for some $c \in (a, b)$. Requires f to be:

a. positive

b. integrable and bounded

c. **continuous** ✓

d. analytic

MVT for integrals: f continuous on $[a, b]$.

42 [Challenging] • Target: Integrals of piecewise functions • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to x?

a. $x^2 - 1$ ✓

b. x^2

c. $x^2 - 1 - (2^2 - 1)$

d. $2x$

The Fundamental Theorem of Calculus, Part 1, states that if $F(x) = \int_a^x f(t)dt$, then $F'(x) = f(x)$. In this problem, $f(t) = t^2 - 1$ and the lower limit of integration is $a = 2$. Therefore, $F'(x) = x^2 - 1$.

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43 [Challenging] • Target: Visualizing the two parts of FTC together • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Mean Value Theorem for Integrals: if f is continuous on $[a, b]$, there exists $c \in (a, b)$ with $\int_a^b f(x)\,dx =$

a. $f(c)(b-a)$ ✓

b. $f'(c)(b-a)$

c. $f(b) - f(a)$

d. $f(c)$

This is the integral form of the MVT, closely tied to FTC Part 1: the average value $f(c)$ times the interval length equals the integral.

44 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 6.01 Related Rates

Find the average value of $f(x) = \cos(x)$ on $[0, \frac{\pi}{2}]$.

a. $\frac{2}{\pi}$ ✓

b. $\frac{4}{\pi}$

c. 1

d. $\frac{1}{\pi}$

Average value $= \frac{1}{\frac{\pi}{2}-0}\int_0^{\frac{\pi}{2}} \cos(x)\,dx = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$.

45 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 6.02 Optimization and Curve Sketching

Net area from 0 to 2π of $\sin x$:

a. -2

b. 0 ✓

c. 2

d. 4

Applying Fundamental Theorem principles, $[-\cos x]_0^{2\pi} = 0$. Therefore, the verified correct option is 0.

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